

Day 3 Extra Homework - Linux Commands and Log Analysis

Level 1: Basic Commands

ls – Lists files and folders in the current directory.

pwd – Shows the current working directory.

cd – Changes the current directory.

cat – Displays file contents.

Task 3: File Creation and Viewing

```
winterfell@winterfell:~/Documents/cyber_security_lex$ touch test.txt
winterfell@winterfell:~/Documents/cyber_security_lex$ cat > test.txt
this is a test file
winterfell@winterfell:~/Documents/cyber_security_lex$ cat test.txt
this is a test file
winterfell@winterfell:~/Documents/cyber_security_lex$
```

touch – Used to create a new file.

cat > filename – Used for writing text into a file.

nano – Text editor used to edit files.

cat filename – Displays the contents of a file.

Level 2: Log Reading

```
winterfell@winterfell:~/Documents/cyber_security_lex$ ls
alternatives.log      boot.log.6           gdm3
alternatives.log.1    boot.log.7           gpu-manager.log
alternatives.log.2.gz bootstrap.log         hp
alternatives.log.3.gz btmnp                installer
alternatives.log.4.gz btmnp.1              journal
alternatives.log.5.gz cloud-init.log        kern.log
apport.log            cloud-init-output.log kern.log.1
apport.log.1          cups                 kern.log.2.gz
apport.log.2.gz       cups-browsed         kern.log.3.gz
apport.log.3.gz       dist-upgrade         kern.log.4.gz
apport.log.4.gz       dmesg                lastlog
apport.log.5.gz       dmesg.0              openvpn
apport.log.6.gz       dmesg.1.gz           private
apport.log.7.gz       dmesg.2.gz           README
apt                   dmesg.3.gz           speech-dispatcher
auth.log              dmesg.4.gz           sssd
auth.log.1            dpkg.log             syslog
auth.log.2.gz         dpkg.log.1           syslog.1
auth.log.3.gz         dpkg.log.2.gz        syslog.2.gz
auth.log.4.gz         dpkg.log.3.gz        syslog.3.gz
boot.log              dpkg.log.4.gz        syslog.4.gz
boot.log.1            dpkg.log.5.gz        sysstat
boot.log.2            dpkg.log.6.gz        unattended-upgrades
boot.log.3            dpkg.log.7.gz        wtmp
boot.log.4            falllog
boot.log.5            fontconfig.log
winterfell@winterfell:~/Documents/cyber_security_lex$
```

Types of logs identified:

Hardware Logs: kernel, USB, CPU, disk, memory, PCI.

System Startup and Shutdown Logs.

Network Logs.

Sample Log Entries

2026-05-14T14:53:56.003083+05:30 winterfell kernel: usb 1-1: USB disconnect, device number 2

2026-05-14T14:53:57.946473+05:30 winterfell kernel: Logitech Wireless Mouse detected.

2026-05-14T15:12:17.598326+05:30 winterfell NetworkManager: Wired connection set as default for IPv4 routing and DNS.

Task 5: Authentication Logs

```

0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-14T15:41:27.002083+05:30 winterfell gdm-password]: message repeated 2 times: [ pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell]
2026-05-14T15:41:32.890773+05:30 winterfell gdm-password]: gkr-pam: unlocked login keyring
2026-05-14T15:45:01.577069+05:30 winterfell CRON[9218]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-05-14T15:45:01.583986+05:30 winterfell CRON[9218]: pam_unix(cron:session): session closed for user root
winterfell@winterfell:~$ grep "authentication failure" /var/log/auth.log
2026-05-08T20:52:59.363024+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-10T22:30:16.915441+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-11T20:37:22.473012+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname= uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-12T14:45:20.842093+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-12T14:45:26.418226+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-13T18:03:32.147058+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-13T18:03:32.147058+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-14T14:53:23.606443+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-14T15:41:18.036491+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-14T15:41:27.002083+05:30 winterfell gdm-password]: message repeated 2 times: [ pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell]
winterfell@winterfell:~$ grep "authentication failure" /var/log/auth.log wc -l
/var/log/auth.log
grep: wc: No such file or directory
winterfell@winterfell:~$ grep "authentication failure" /var/log/auth.log | wc -l

```

gkr-pam: unlocked login keyring

pam_unix(gdm-password:auth): authentication failure for user winterfell

Number of failed login attempts detected: 9

Task 7 & 8: Suspicious Activity Detection

```

0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-14T15:41:27.002083+05:30 winterfell gdm-password]: message repeated 2 times: [ pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell]
2026-05-14T15:41:32.890773+05:30 winterfell gdm-password]: gkr-pam: unlocked login keyring
2026-05-14T15:45:01.577069+05:30 winterfell CRON[9218]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-05-14T15:45:01.583986+05:30 winterfell CRON[9218]: pam_unix(cron:session): session closed for user root
winterfell@winterfell:~$ grep "authentication failure" /var/log/auth.log
2026-05-08T20:52:59.363024+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-10T22:30:16.915441+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-11T20:37:22.473012+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname= uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-12T14:45:20.842093+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-12T14:45:26.418226+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-13T18:03:32.147058+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-14T14:53:23.606443+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-14T15:41:18.036491+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-14T15:41:27.002083+05:30 winterfell gdm-password]: message repeated 2 times: [ pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell]
winterfell@winterfell:~$ grep "authentication failure" /var/log/auth.log wc -l
/var/log/auth.log
grep: wc: No such file or directory
winterfell@winterfell:~$ grep "authentication failure" /var/log/auth.log | wc -l

```

Multiple authentication failures were detected across several dates. This may indicate repeated wrong password attempts or a possible brute-force attack targeting the user account 'winterfell'. The repeated failures occurring in a short period make the activity suspicious.

Suspicious Indicators:

- Multiple failed authentication attempts.
- Repeated login failures over several days.
- The same user account was targeted repeatedly.
- Events occurred within a short time period.

Task 9 creating Your own fake log

```

2026-05-14T16:05:01.610295+05:30 winterfell CRON[11085]: pam_unix(cron:session): session closed for user root
2026-05-14T16:13:36.090201+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-14T16:13:41.411345+05:30 winterfell gdm-password]: pam_unix(gdm-password:auth): authentication failure; logname=winterfell uid=0 euid=0 tty=/dev/tty1 ruser= rhost= user=winterfell
2026-05-14T16:13:48.043064+05:30 winterfell gdm-password]: gkr-pam: unlocked login keyring

```

Task 10: Failed Login Attempt Analysis

If a system shows 50 failed login attempts within one minute, it may indicate a brute-force attack. In a brute-force attack, an attacker repeatedly tries different passwords to gain unauthorized access. Such activity is considered suspicious because normal users do not usually fail authentication many times in a short duration. Security analysts should investigate the source IP addresses, monitor authentication logs, and enable additional security measures such as multi-factor authentication.